

Inventory Subsidiary Ledgers

⇒ A separate subsidiary account is maintained for each item in inventory.

Perpetual inventory Methods

- FIFO
- LIFO
- Weighted Average

FIFO

⇒ Assumes materials received first in the stores are first to be sold. materials in stock are the materials purchased last.

⇒ Inventory items are issued/sold at the oldest price listed in the stores ledgers until the first batch fully utilized.

in inflationary environments, FIFO results in a higher ending inventory value and lower COGS, leading to higher net income. This method is commonly used when inventory costs are rising.

Advantages:

- Simple and easy
- Closing stock value nearer to current price market.

Disadvantages:

- improper matching of cost with revenue since the cost of goods sold is computed on the basis of old prices.
- Difficult to compare two jobs.

Example:

Aug 01: Balance or beginning inventory 1000 units, \$10 each.

Aug 03: Purchased 4,000 units @ \$12 each

Aug 10: Sold 3,000 units @ \$20 each.

Aug 17: Purchased 6,000 units @ \$15 each.

Aug 25: Sold 5,000 units @ \$20 each.

Date	Purchases	Sales	Ending Balance
Aug. 01			1000 @ 10 = 10,000
03	4000 @ 12 = 48,000		1000 @ 10 = 10,000 4000 @ 12 = 48,000
10		1000 @ 10 = 10,000 2000 @ 12 = 24,000	2000 @ 12 = 24,000
17	6000 @ 15 = 90,000		2000 @ 12 = 24,000 6000 @ 15 = 90,000
15		2000 @ 12 = 24,000 3000 @ 15 = 45,000	3000 @ 15 = 45,000
	10000 138,000	8000 103,000	3000 45,000

- Total purchases = 138,000
- Cost of goods sold = 103,000
- Cost of ending inventories = 45,000

$$\therefore \text{OI} + \text{Purchases} - \text{COGS} = \text{EI}$$

$$10,000 + 138,000 - 103,000 = 45,000$$

$$45,000 = 45,000$$

LIFO Method

نظام لIFO

- Assumes that the material or goods received last, in the stores are the first to be sold.

used to reduce taxable income during periods of inflation.

Advantage:

- Realistic reported profit
- Tax benefit & improvement in CF.
- LIFO minimizes the write downs to market.

Disadvantages:

- Unrealistic values of inventories
- Reduced earnings in inflationary times
- Understatement of inventory

Example:

(Statement on previous Page)

Date	Purchases	Sales	Balance
01			1000 @ 10 = 10,000
03	4000 @ 12 = 48,000		1000 @ 10 = 10,000 4000 @ 12 = 48,000
10		3000 @ 12 = 36,000 2000 @ 12 = 24,000	1000 @ 10 = 10,000 1000 @ 12 = 12,000
17	6000 @ 15 = 90,000		1000 @ 10 = 10,000 2000 @ 12 = 24,000 6000 @ 15 = 90,000
15		5000 @ 15 = 75,000	1000 @ 10 = 10,000 1000 @ 12 = 12,000 1000 @ 15 = 15,000 3000 37,000
	10000 138,000	8000 103,000	3000 37,000

$$\begin{array}{rcl}
 \text{Sales} & = & 8000 \times 20 \quad 160,000 \\
 \text{Less} & = & \text{COGS} \quad (111,000) \\
 & & \hline
 & & 49,000
 \end{array}$$

(Average Cost Methods)

⇒ Assumes that the goods available for sale are homogeneous and values inventory on the basis of weighted average price. Calculates the average cost of all inventory items and assigns that cost to each unit sold. Computed after every purchase.

Pros:

- more scientific and smoothes the fluctuation in purchase price.

- Inventory is valued at one date. Suitable for homogeneous inventory system.

Cons:

- Tedious method.
- Issue price need to be computed at each purchase.

Purchases:

- Balance units + purchase unit
- Balance total cost + purchase total cost
- Unit cost: Total cost / Total units (change after every purchase)

Sales:

- Balance units - Sales units (Less)
- Balance total cost - Sales total cost (Less)
- unit cost: Remains constant

Date	Purchases	Sales	Balance
Aug 01			1000 @ 10 = 10000
Aug 03	4000 @ 12 = 48000		5000 @ 11.60 = 58000
Aug 10		3000 @ 11.60 = 34800	2000 @ 11.60 = 23200
Aug 17	6000 @ 15 = 90000		8000 @ 14.15 = 113200
Aug 25		5000 @ 14.15 = 70750	3000 @ 14.15 = 42450
	10,000 138,000	8000 105,550	3000 42450

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EX-S-2

(a)

Date	Detail	Dr \$	Cr \$
may 10	Inventory (540x5)	2700	
	A/c payable		2700
may 23	A/c Rec. (900x4)	3600	
	Sales		3600
may 23	Cost of sales (540x4)	2160	
	Inventory		2160
	A/c payable	2700	
	Cash		2700
may 29			
june 19	Cash (950x2)	1900	
	Sales		1900
june 19	Cost of sales (540x2)	1080	
	Inventory		1080
june 22	Cash	3600	
	A/c Rec.		3600

COGS = 3217130
Inventory 22950 +
Inventory 11710 +
3251850

Page #265 (Problems)

(b) Sales: 4966000
Less COGS (3251850)
Gross Profit 1714150

Date	Detail	Dr \$	Cr \$
NOV 5	A/c Rec.	38400	
	Sales		38400
	Cost of sales	22950	
	Inventory		22950
NOV 9	Inventory	104000	
	A/c payable		104000
	Cash	38400	
DEC 5	A/c Rec.		38400
DEC 9	A/c payable	104000	
	Cash		104000
DEC 31	COGS inventory (975130 - 964360)	10770	10770

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(S-10 - EX)

Riviera Fashion → seller
Carolines → Buyer

Discount term 2/10, n/30
After Discount period?
Within Discount period?

(a) Riviera fashion

Date	Detail	Dr \$	Cr \$
Mar 3	A/c Rec. (50 x 1000)	50000	
	Sales		50000
	Cost of sales (32 x 1000)	32000	
	Inventory		32000
Mar 5	Sales return (100 x 50)	5000	
	A/c Rec.		5000
	Inventory (100 x 32)	32000	
	Cost of sales		32000
	Cash	44100	
Mar 13	Sales discount	900	
	A/c Rec.		45000

(b) Caroline - mid cos

Date	Detail	Dr \$	Cr \$
Mar 3	Inventory (50000 x 2/100)	49000	
	A/c payable		49000
Mar 5	A/c payable (50000 x 2/100)	4900	
	Inventory		4900
	A/c payable	44100	
Mar 13	Cash		44100

(i) Riviera fashion (Last entry)
Cash 45000

A/c Rec. 45000

Caroline (Last entry)

A/c payable 44100

Discount loss 900

Cash 45000

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Date	Detail	Dr (\$)	Cr (\$)
Jan 3	Cash	18,900	
	Sales		18,900
	Cost of sales	11200	
	Inventory		11200
	Inventory	9600	
Jan 6	A/c payable		9600

Inventory Periodic

Date	Detail	Dr \$	Cr \$
Jan 3	Cash	18900	
	Sales		18900
Jan 6	Purchase	9600	
	A/c payable		9600

(b) Balance = ?
ending inventory = Balance

ii Open inventory + purchases - EI = COGS
2200 + 9600 - 20400 = 11200

(d)

COGS = 20400

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(EX - S-S)
(a) Perpetual

Date	Detail	Dr \$	Cr \$
jan 2	Cash	40000	
	Sales		40000
	Cost of Sales	28,000	
	Inventory		28,000
jan 5	Inventory	18,500	
	A/c payable		18,500

(c) Periodic

Date	Detail	Dr \$	Cr \$
jan 2	Cash	40,000	
	Sales		40,000
jan 5	Purchase	18,500	
	A/c payable		18,500

(b)

OI + Purchases - EI = COGS
90000 + 18500 - 80500 = 28000

(c)

OI + Purchases - EI = COGS
90000 + 18500 - 80500 = 28000

Chapter #9

Inventory Systems

Two types:

1. Perpetual system
2. Periodic system

Inventory Methods

→ Three main types:

1. FIFO
2. LIFO
3. Average Cost Method

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FIFO:

Cost of sale

			Total cost
Oct 1	2	\$3000	\$6000
Nov 7	2	32,00	6400
	4		12400

Cost of sale 12400
Inventory 12400

Ending inventory:

			Total cost
Nov 17	1	\$3200	\$3200
Dec 1	4	3250	13000
			16200
Nov 17	3	\$3200	\$9600
Oct 1	1	3000	3000
			12600

Cost of sale 12600
Inventory 12600

Ending inventory

			Total cost
Oct 1	1	\$3000	\$3000
Dec 1	4	3250	13000
			16000

Average cost method:
Per unit cost = $\frac{15600}{5} = 3120$

Cost of sale (4 x 3120) 12480
Inventory 12180

Sold 4 Audio phone on Dec 5

FIFO:

Nov 17	2	\$ 3000	Total cost
	2	3200	\$6000
	4	6200	6400
			12400

Nov 17	1	\$ 3200	Ending inventory
Dec 1	4	3250	Total cost
	5		\$ 3200
			13000
			16200

LIFO:

Dec 1	4	\$ 3250	Total cost
	4		\$ 13000
			13000

Cost of sale 13000

Oct 1	2	\$ 3000	Inventory
Nov 17	3	3200	Ending inventory
			Total cost
			\$ 6000
			9600
			15600

Average:

$$\text{Per unit cost} = 28600 / 9 = 3177.8$$

Cost of sale (4 x 3177.8) 27111.1

Inventory	27111.1
Ending inventory	

$$5 \times 3177.8 = 15889$$

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(1) FIFO:

Cost of sale

April 9	60	\$ 800	Total cost
April 1	10	850	\$48000
			8500
			56500

May 1	30	\$ 850	Ending inventory
	30		Total cost
			\$25500
			25500

(2) LIFO:

Cost of sale

May 1	40	\$ 850	Total cost
April 9	30	800	\$ 34000
	70		24000
			58000

Cost of sale 58000

Inventory	58000
<u>Ending inventory</u>	
	Total cost
<u>30</u>	\$ 800
30	<u>2400</u>
	2400

Average cost Method

cost of sale

$$\text{Per unit cost} = 82000 / 100 = 820$$

$$\text{cost of sale (820 x 70)} = 57400$$

	Ending inventory	Total cost
May 1	30	\$ 820
	30	\$24600
	30	24600

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Ending inventory units = 1500

Total units = 10,000

Sold units = 10,000 - 1500

= 8500

Perpetual

FIFO:

			Total cost
Jan 1	900	\$ 10.00	\$ 9000
Feb 23	1200	11.00	13,200
Apr 20	3000	11.20	33,600
May 4	3700	11.60	39,440
			95240

cost of sale 95240
Inventory 95240

	Ending inventory	Total cost
May 4	600 \$ 11.60	\$ 6960
Nov 30	900 13.00	11700
	1500	18660

LIFO:

			Total cost
Nov 30	900	\$ 13.00	\$ 11700
May 4	4000	11.60	46400
Apr 20	3000	11.20	33600
Feb 23	600	11.00	6600
	8500		98300

	Ending inventory	Total cost
Feb 23	600 \$ 11.00	\$ 6600
Jan 1	900 10.00	9000
	1500	15600

Average cost:

$$\text{Per unit cost} = 113,900 / 10,000 = 11.39$$

$$\text{cost of sale (8500 x 11.39)} = 96815$$

	Inventory	Total cost
	Ending inventory	96815
1500	\$ 11.39	\$ 17085
		17085

2. find out ending inventory periodic

Periodic

FIFO:

	Ending inventory	Total cost
Nov 30	900	\$ 13.00 \$ 11700
May 4	600	11.60 69600
	<u>1500</u>	<u>18660</u>

Cost of goods available for sale	113900
Less ending inventory	(18660)
Cost of sale	<u>95240</u>

LIFO:

	Ending inventory	Total cost
Jan 1	900	\$ 10.00 \$ 9000
Feb 23	600	\$ 11.00 6600
	<u>1500</u>	<u>15600</u>

Cost of goods available for sale	113900
Less ending inventory	(15600)
Cost of sale	<u>98300</u>

Average cost Method

Per unit cost = $113900 / 10,000 = \$11.39$

Ending inventory

$1500 \times 11.39 = \$17085$

Cost of goods available for sale	113900
Less ending inventory	(17085)
Cost of sale	<u>\$ 96815</u>

Past Paper - 2023

Q3:

(a)

Perpetual system

FIFO:

	Cost of Sale	Total cost
Apr 1	4 \$ 8000	\$ 32000
Apr 19	1 8200	8200
	<u>5</u>	<u>40200</u>
Cost of sale	40200	
Inventory		<u>40200</u>
Ending inventory		
Apr 19	4 \$ 8200	32800
May 8	3 8500	25500
	<u>5</u>	<u>58300</u>

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(Exercise - 9-9)

FIFO:

Ending inventory

			Total cost
Third purchase (Aug 24)	3100	\$4.60	14260
Fourth purchase (Aug 30)	10900	5.00	54,500
			<u>\$68760</u>

Cost of goods available for sale	530,400
less ending inventory	(68760)
<u>Cost of sale</u>	<u>461640</u>

LIFO:

Ending inventory

			Total cost
Jan 1	9100	\$4.00	\$36400
Firs Purchase (Feb 20)	49,000	4.10	<u>20090</u>
			<u>\$56490</u>

Cost of goods available for sale	530,400
less ending inventory	(56490)
<u>Cost of sale</u>	<u>\$473910</u>

Average cost:

Ending inventory

Cost per unit = $\$530400 / 120,000$

Cost per unit = 4.42

EI	114000	4.42	Total cost
			\$61880
			<u>\$61880</u>

Cost of goods available for sale 53040
 Less ending inventory (61880)
 Cost of sale \$468520

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(Problem 9-2)

Date	Purchase			Sale			Balance	
	Units	Unit Cost	Total	Units	Unit Cost	Total	Units	Unit Cost
May 1							20 @ 40	800
May 5				8 @ 40	320		12 @ 40	480
May 6	20 @ 45		900				12 @ 40	480
May 21				10 @ 45	450		20 @ 45	900
May 31				10 @ 45	450		12 @ 40	480
				5 @ 40	200		10 @ 45	450
Total	20		900	33	1420		7	280

(Problem 9-3)

Date	Purchase			Sale			Balance	
	Units	Unit Cost	Total	Units	Unit Cost	Total	Units	Unit Cost
Sep 1							50 @ 60	3000
Sep 4	20 @ 65		1300				50 @ 60	3000
Sep 8				35 @ 60	2100		20 @ 65	1300
Sep 9	40 @ 65		2600				15 @ 60	900
							20 @ 65	1300
							40 @ 65	2600

Sep 20			15 @ 60	900	15 @ 65	975
			20 @ 65	1300		
			25 @ 65	1625		
Sep 25	40 @ 70	2800			15 @ 65	975
					40 @ 70	2800
Sep 30			5 @ 65	325	10 @ 65	650
					40 @ 70	2800
Total	100	6700	100	6250	50	3450

• Total purchases = 6700

• Cost of goods sold = 6250

• Cost of ending inventories = 3450

Past Papers - 2023

(a) (Perpetual System)
FIFO

		Cost of Sale	Total Cost
Apr 1	4	\$8000	\$32,000
Apr 19	1	\$8200	8200
			<u>\$40200</u>

Cost of sale 40200
 Inventory 40200

		<u>Ending inventory</u>	<u>Total cost</u>
Apr 19	4	\$8200	\$32800
May 8	3	8500	25500
	7		<u>\$58300</u>

LIFO

Cost of sale

			<u>Total cost</u>
Apr 19	5	\$8200	\$41000
			<u>\$41000</u>

Cost of sale	41000
Inventory	41000

		<u>Ending inventory</u>	<u>Total cost</u>
Apr 1	4	\$8000	\$32000
May 8	3	8500	25500
			<u>\$57500</u>

Average cost

Cost of goods sold till Apr. 19
(32000 + 41000)
= 73,000

Cost per unit = $73,000 / 9 = \$8111.11$

Cost of goods sold ($5 \times \$8111.11$)
= 40555.55

Ending inventory

Remaining units after sale of Apr 28
= 4 unit @ \$8111.11
= 32444.44

Units purchased on May 8 (3 @ 8,500)
= 25500

Inventory = $32444.44 + 25500 = 57944.44$

(b) Periodic System

FIFO

		<u>Ending inventory</u>	<u>Total cost</u>
Apr. 19	4	\$8200	\$32800
May 8	3	8500	25500
			<u>\$58300</u>

Cost of goods available for sale \$98500
Less ending inventory (\$58300)
Cost of sale \$40200

LIFO

	Ending inventory	Total cost
Apr 1	4 \$8000	\$32000
Apr 19	3 8200	24600
		<u>\$56600</u>

Cost of goods available for sale \$98,500
 Less ending inventory (\$56600)
 Cost of sale \$41900

Average Cost Method

$$\text{Cost per unit} = 98,500 / 12 = 8208.33$$

$$\text{Ending inventory} = 7 @ 8208.33 = 57458.33$$

Cost of goods available for sale \$98,500
 Less ending inventory (\$57458.33)
 Cost of sale \$41041.65

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 (EX-9-4)

FIFO

	Cost of goods sold	Total cost
Dec 12	1200 \$18.50	\$22200
Jan 9	800 19.00	15200
		<u>\$37400</u>

Cost of sale 37400

Inventory 37400
Ending inventory

	Total cost
Jan 9	81000 \$19.00 \$19000
	<u>\$19000</u>

LIFO

	Cost of goods sold	Total cost
Jan 9	1800 \$19.00	\$34,200
Dec 12	200 18.50	3700
		<u>\$37900</u>

Ending inventory

	Total cost
Dec 12	1000 \$18.50 \$18500
	<u>\$18500</u>

Average Cost

$$\text{Cost per unit} = \$56,400 / 3000 = 188$$

$$\text{Cost of goods sold} (2000 \times 188) = 376,000$$

Inventory
Ending Inventory

376,000

$$= 1000 @ 188$$

$$\text{ending inventory} = \$188,000$$